

LangGraph Course Book

Module 1: Introduction to LangGraph

- Graphs vs Chains
- State Machines
- Execution Model

Module 2: Core Components

- Nodes
- Edges
- State
- Reducers
- Messages

Module 3: Building Graphs

- StateGraph
- Entry Points
- Finish Points
- Compilation

Module 4: State Management

- TypedDict
- Pydantic State
- Reducers
- Updates

Module 5: Routing

- Conditional Edges
- Branching
- Dynamic Paths

Module 6: Human-in-the-Loop

- Interrupts
- Approvals
- Feedback Loops

Module 7: Tool Integration

- Tool Nodes
- Function Calling

- External APIs

Module 8: Persistence

- Checkpointing
- Thread State
- Recovery

Module 9: Agent Workflows

- ReAct
- Planner-Executor
- Reflection Agents

Module 10: Multi-Agent Graphs

- Supervisor Pattern
- Agent Teams
- Coordination

Module 11: Streaming & Debugging

- Streaming Events
- Tracing
- LangSmith

Module 12: Deployment

- Scaling
- Cloud Deployment
- Monitoring

Module 13: Advanced LangGraph

- Graph RAG
- Recursive Graphs
- Agent Swarms

Module 14: Capstone Projects

- Research Assistant
- Enterprise Automation
- Multi-Agent Platform